



Universität  
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# Resolving the Remaining Cases of Carlitz-Extensiveness over Finite Fields using SageMath

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# Main goal of this talk

## The question

For which finite field extensions  $\mathbb{F}_{q^n}/\mathbb{F}_q$  does there exist, for every  $z \in \mathbb{F}_{q^n}$ , an element  $x \in \mathbb{F}_{q^n}$  that is both **primitive** and a  **$\gamma_z$ -generator**?

- Open since Hsu & Nan (2011); narrowed by Hachenberger (2016).
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- **Six pairs**  $(q, n)$  remained unresolved.

## What we show

Using SageMath, we settle the six remaining cases. The complete classification:

$$\mathbb{F}_{q^n}/\mathbb{F}_q \text{ is Carlitz-extensive} \iff (q, n) \neq (2, 2).$$

# The Primitive Normal Basis Theorem

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# Primitive and normal elements

Let  $\mathbb{F}_{q^n}/\mathbb{F}_q$  be a finite field extension of degree  $n$ .

## Primitive element ( $\rightarrow$ multiplicative structure)

$\alpha \in \mathbb{F}_{q^n}$  is **primitive** if  $\mathbb{F}_{q^n}^\times = \langle \alpha \rangle$ . Equivalently,  $\text{ord}(\alpha) = q^n - 1$ .

## Normal element ( $\rightarrow$ additive structure)

$\alpha \in \mathbb{F}_{q^n}$  is **normal** over  $\mathbb{F}_q$  if the iterates of the Frobenius automorphism form an  $\mathbb{F}_q$ -basis of  $\mathbb{F}_{q^n}$ :

$$\{\alpha, \alpha^q, \alpha^{q^2}, \dots, \alpha^{q^{n-1}}\} \text{ is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

# The Primitive Normal Basis Theorem (PNBT)

## Theorem (Lenstra & Schoof, 1987)

For every finite field extension  $\mathbb{F}_{q^n}/\mathbb{F}_q$  there exists an element  $\alpha \in \mathbb{F}_{q^n}$  which is simultaneously primitive and normal over  $\mathbb{F}_q$ .

Such an  $\alpha$  generates  $\mathbb{F}_{q^n}^\times$  multiplicatively and  $\mathbb{F}_{q^n}$  additively (via its Frobenius basis).

Lenstra–Schoof

Hsu & Nan

Hachenberger

Gruber

this work

## The Hsu–Nan Generalisation (2011)

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# The endomorphism $\gamma_z$

Hsu & Nan (2011) generalise the Primitive Normal Basis Theorem by replacing the Frobenius automorphism  $\sigma$  with a wider family of  $\mathbb{F}_q$ -linear maps.

## Definition

For  $z \in \mathbb{F}_{q^n}$ , define

$$\gamma_z: \mathbb{F}_{q^n} \longrightarrow \mathbb{F}_{q^n}, \quad \gamma_z(x) = z \cdot x + x^q.$$

- $\gamma_z$  is an  $\mathbb{F}_q$ -linear endomorphism of  $\mathbb{F}_{q^n}$ .
- **Special case  $z = 0$ :**  $\gamma_0(x) = x^q = \sigma(x)$  is the Frobenius automorphism.

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## Lemma (Minimal polynomial, Hsu–Nan)

If  $z \in \mathbb{F}_{q^n}$  has minimal polynomial  $f_z(x) \in \mathbb{F}_q[x]$  of degree  $d \mid n$ , then

$$m_{\gamma_z}(x) = f_z(x)^{n/d} - 1.$$

## Definition ( $\gamma_z$ -generator)

An element  $\alpha \in \mathbb{F}_{q^n}$  is a  $\gamma_z$ -**generator** if

$$\{\alpha, \gamma_z(\alpha), \gamma_z^2(\alpha), \dots, \gamma_z^{n-1}(\alpha)\} \quad \text{is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

- A  $\gamma_z$ -generator **always exists** but does **not need to be primitive**.

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- A  $\gamma_z$ -generator **always exists** but does **not need to be primitive**.

## Definition (Carlitz-extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$  is **Carlitz-extensive** if for every  $z \in \mathbb{F}_{q^n}$  there exists a primitive  $\gamma_z$ -generator.

# Hsu & Nan: open cases

Hsu & Nan worked with character-sum estimates and solved all but 63 pairs  $(q, n)$  pairs. These 63 pairs were too small for asymptotic methods to apply.

## Known exception

$(q, n) = (2, 2)$ , i.e.  $\mathbb{F}_4/\mathbb{F}_2$ , is *not* Carlitz-extensive. Is it the only one?

*Hsu & Nan, J. Number Theory 131 (2011), 146–157.*

(Subsequent work reduced the 63 open cases to just six — as we will see.)



# The Hachenberger Generalisation (2016)

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# Hachenberger: cyclic vector spaces

Hachenberger (2016) embeds the problem in a wider framework: replace  $\gamma_z$  by *any*  $\mathbb{F}_q$ -linear endomorphism  $\tau$  of  $\mathbb{F}_{q^n}$  for which a  $\tau$ -generator exists (analogue to  $\gamma_z$ -generator). Such a  $\tau$  is then called *cyclic*.

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## Definition (Extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$  is **extensive** if for every cyclic  $\mathbb{F}_q$ -endomorphism  $\tau$  of  $\mathbb{F}_{q^n}$  there exists a primitive  $\tau$ -generator  $\alpha \in \mathbb{F}_{q^n}$ , i.e.  $\alpha$  generates both  $\mathbb{F}_{q^n}^\times$  and  $\mathbb{F}_{q^n}$  as an  $\mathbb{F}_q[\tau]$ -module.

- Hachenberger solved the question of extensivity for all but 19 pairs  $(q, n)$ .
- **Specialisation:**  $\tau = \gamma_z, z \in \mathbb{F}_{q^n} \Rightarrow$  primitive  $\gamma_z$ -generator  $\rightarrow$  Hsu–Nan

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Progress on the 63 open cases:

- Hachenberger resolved 41 of them.
- Master's thesis of Thomas Gruber reduced the list by another 16 pairs.

Hachenberger, *Australasian J. Combin.* 64 (2016), 289–326.

F. Hellwig & M. Schaller

Lenstra–Schoof

Hsu & Nan

Hachenberger

Gruber

this work

1987

2011

2016

2016

2026

# The remaining 6 cases

After Hachenberger and Gruber, only **six pairs**  $(q, n)$  remained unresolved:

| $q$ | $n$        |
|-----|------------|
| 2   | 18, 20, 24 |
| 4   | 10, 12     |
| 8   | 8          |

- **Too large** for naive brute force.
- **Too small** for asymptotic / structural arguments.

**Goal of this work:** settle each case by an explicit computation in SageMath.



## Our Approach: Reductions & Algorithm

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# Strategy at a glance

We reduce the search using:

1. **Cyclotomic cosets**: one  $z$  per orbit (corresponding to coset) suffices (outer loop), simplifies the calculation of the minimal polynomial  $f_z$  of  $z$ .

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1. **Cyclotomic cosets**: one  $z$  per orbit (corresponding to coset) suffices (outer loop), simplifies the calculation of the minimal polynomial  $f_z$  of  $z$ .
2. **Projective space**: one  $x$  per projective line suffices (inner loop), restrict the search for a primitive  $\gamma_z$ -generator to one representative per projective primitive line.
3. **Minimal-polynomial factorisation**: fast algebraic test for the  $\gamma_z$ -generator property, more efficient than e.g. the basis check.

Each tool decreases the number of computations needed, making the algorithm feasible.

# The algorithm in one slide

**Input.** A pair  $(q, n)$ . **Output.** True / False for Carlitz-extensiveness.

## Preparation.

1. Fix a primitive element  $g \in \mathbb{F}_{q^n}^\times$ ; set  $N = q^n - 1$ ,  $Q = N/(q - 1)$ .

2. Precompute representatives of the primitive projective points:

$$\mathcal{R} = \{g^k : 1 \leq k < Q, \gcd(k, Q) = 1\}.$$

3. Handle  $z = 0$  separately (Frobenius case): search  $\mathcal{R}$  for a primitive  $\gamma_0$ -generator.

## Main loop. For each $q$ -cyclotomic coset of $\mathbb{Z}/N\mathbb{Z}$ :

1. Pick representative  $z = g^e$  of Frobenius orbit.
2. Compute the minimal polynomial  $f_z$  of  $z$  via its cyclotomic coset, then factor  $m_{\gamma_z} = f_z^{n/\deg f_z} - 1 = \prod_i p_i^{e_i}$ .
3. Build the matrix  $A$  of  $\gamma_z$ , then  $H_i = (m_{\gamma_z}/p_i)(A)$  for each  $i$ .
4. Scan  $\mathcal{R}$ : if  $\text{Vec}(g^k) \cdot H_i \neq 0$  for all  $i$ , then  $\alpha = \lambda g^k$  is a primitive  $\gamma_z$ -generator.
5. If  $\mathcal{R}$  is exhausted with no hit: return False.

If every coset succeeds: return True.

## Results

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# Main result

## Theorem (Hellwig & Schaller, 2026)

*Each of the six remaining pairs*

$$(q, n) \in \{(2, 18), (2, 20), (2, 24), (4, 10), (4, 12), (8, 8)\}$$

*is Carlitz-extensive.*

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is *Carlitz-extensive*.

## Combined classification

A finite-field extension  $\mathbb{F}_{q^n}/\mathbb{F}_q$  is Carlitz-extensive *if and only if*  $(q, n) \neq (2, 2)$ .

Equivalently,  $\mathbb{F}_4/\mathbb{F}_2$  is the *unique* non-Carlitz-extensive extension among all finite extensions of finite fields.

## Two independent searches

- **Original algorithm** (projective-point first):  $\approx 18\text{h}$  on one core for all six cases together.
- **Alternative algorithm** (generator-first): find one  $\gamma_z$ -generator  $u$ , then search candidates  $a(\gamma_z)(u)$  with  $a \in \mathbb{F}_q[x]$  monic,  $\deg a < n$ ,  $\gcd(a, m_{\gamma_z}) = 1$ .  $\approx 21\text{h}$  on one core.

Both produce identical results on all six cases.

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## Repository

Code, pseudocode, benchmarks, and additional material:

`github.com/florian-hellwig/Carlitz-extensiveness`

## Outlook: Generalisations and Conjectures

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# $\mathbb{F}_q[x]$ -Drinfeld extensiveness: an intermediate framework

Let  $\tau: x \mapsto x^q$  be the Frobenius on  $\mathbb{F}_{q^n}$ .

## Definition ( $\mathbb{F}_q[x]$ -Drinfeld endomorphism of rank $r$ )

For  $t, c_1, \dots, c_r \in \mathbb{F}_{q^n}$  with  $c_r \neq 0$ ,

$$\varphi = t \cdot \tau^0 + c_1 \tau + \dots + c_r \tau^r \in \text{End}_{\mathbb{F}_q}(\mathbb{F}_{q^n}), \quad \varphi(v) = tv + c_1 v^q + \dots + c_r v^{q^r}.$$

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## Definition ( $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank $\leq r$ )

$\mathbb{F}_{q^n}/\mathbb{F}_q$  is  $\mathbb{F}_q[x]$ -**Drinfeld-extensive of rank  $\leq r$**  if for every *cyclic*  $\mathbb{F}_q[x]$ -Drinfeld endomorphism  $\varphi$  of rank  $\leq r$ , there exists a primitive  $\varphi$ -generator  $\alpha \in \mathbb{F}_{q^n}$ .

- Any  $\mathbb{F}_q[x]$ -Drinfeld endomorphism of rank  $r = 1$  is *cyclic*. Not the case for  $r \geq 2$ .
- $r \leq 1$  with  $c_1 = 1 \implies \varphi = \gamma_t$  & *Carlitz-extensiveness*
- $r \geq n - 1$ : Any  $\mathbb{F}_q$ -linear map is  $\mathbb{F}_q[x]$ -Drinfeld polynomial  $\rightarrow$  Hachenberger-*extensiveness*

# $\mathbb{F}_q[x]$ -Drinfeld-Extensiveness: Classification

## Hierarchy

Extensive  $\implies \mathbb{F}_q[x]$ -Drinfeld-extensive of rank  $\leq r$  for every  $r \geq 1$ .

### Known non-extensive pairs:

| $(q, n)$                 | $\mathbb{F}_q[x]$ -Drinfeld status                                                       |
|--------------------------|------------------------------------------------------------------------------------------|
| $(2, 2)$                 | Not Carlitz-ext. $\Rightarrow$ <b>not</b> $\mathbb{F}_q[x]$ -Drinfeld-ext. rank $\leq 1$ |
| $(3, 2), (5, 2), (2, 4)$ | Carlitz-ext., <b>not</b> $\mathbb{F}_q[x]$ -Drinfeld-ext. rank $\leq 1$                  |
| $(2, 6)$                 | $\mathbb{F}_q[x]$ -Drinfeld-ext. rank $\leq 2$ , <b>not</b> rank $\leq 3$                |

### Extensiveness unknown:

$(2, 8), (2, 10), (2, 12), (2, 14),$   
 $(2, 15), (2, 16), (2, 18), (2, 20),$   
 $(2, 24), (3, 8), (3, 10), (3, 12),$   
 $(4, 6), (4, 9), (4, 10), (4, 12),$   
 $(5, 4), (7, 6), (8, 8).$

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$(2, 8), (2, 10), (2, 12), (2, 14), (2, 15), (2, 16), (2, 18), (2, 20), (2, 24), (3, 8), (3, 10), (3, 12), (4, 6), (4, 9), (4, 10), (4, 12), (5, 4), (7, 6), (8, 8).$

## Result (Hellwig & Schaller, 2026)

We prove  $(5, 4)$  is **extensive**, hence  $\mathbb{F}_q[x]$ -Drinfeld-extensive of every rank.

All remaining cases are **conjectured extensive**.

# Simultaneous extensiveness: matching every subfield level

**New question:** Find one primitive  $\alpha \in \mathbb{F}_{q^n}$  that is a generator simultaneously at every intermediate field  $\mathbb{F}_q \subseteq \mathbb{F}_{q^d} \subseteq \mathbb{F}_{q^n}$ ,  $d \mid n$ .

Note that for  $d = n$ , any  $\alpha \neq 0$  is trivially a generator.

- **Simultaneous Carlitz-extensiveness of  $\mathbb{F}_{q^n}/\mathbb{F}_q$ :** for any  $z \in \mathbb{F}_{q^n}$  for which a simultaneous  $\gamma_z^{(d)}$ -generator over  $\mathbb{F}_{q^d}$  exists for all  $d \mid n$ ,  $d < n$ , where  $\gamma_z^{(d)}(v) = zv + v^{q^d}$ , there exists a *primitive* simultaneous  $\gamma_z^{(d)}$ -generator.

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- **Simultaneous  $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank  $\leq r$ :** for any  $t, c_1, \dots, c_r \in \mathbb{F}_{q^n}$  for which a simultaneous  $\varphi^{(d)}$ -generator over  $\mathbb{F}_{q^d}$  exists for all  $d \mid n$ ,  $d < n$ , where  $\varphi^{(d)}(v) = tv + c_1 v^{q^d} + \dots + c_r v^{q^{rd}}$ , there exists a *primitive* simultaneous  $\varphi^{(d)}$ -generator.

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- **Simultaneous  $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank  $\leq r$ :** for any  $t, c_1, \dots, c_r \in \mathbb{F}_{q^n}$  for which a simultaneous  $\varphi^{(d)}$ -generator over  $\mathbb{F}_{q^d}$  exists for all  $d \mid n$ ,  $d < n$ , where  $\varphi^{(d)}(v) = tv + c_1 v^{q^d} + \dots + c_r v^{q^{rd}}$ , there exists a *primitive* simultaneous  $\varphi^{(d)}$ -generator.
- **Simultaneous extensiveness:** for any family  $(\tau_d)_{d \mid n, d < n}$  of  $\mathbb{F}_{q^d}$ -linear endomorphisms of  $\mathbb{F}_{q^n}$  admitting a simultaneous generator, there exists a *primitive* simultaneous generator.

Equivalently: Simultaneous  $\mathbb{F}_q[x]$ -Drinfeld-extens. with  $r \geq n - 1$  and coeffs not fixed.

Fixing  $\varphi^{(d)}(v) = \sigma^d(v) = v^{q^d}$  recovers the **Morgan–Mullen conjecture**.

# Three conjectures on simultaneous extensiveness

## Conjecture (Simultaneous Carlitz-extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$  is simultaneously Carlitz-extensive for *all but finitely many* pairs  $(q, n)$ .

## Conjecture (Simultaneous $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank $\leq r$ )

For each fixed  $r \geq 1$ ,  $\mathbb{F}_{q^n}/\mathbb{F}_q$  is simultaneously  $\mathbb{F}_q[x]$ -Drinfeld-extensive of rank  $\leq r$  for *all but finitely many* pairs  $(q, n)$ .

## Conjecture (Simultaneous extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$  is simultaneously extensive for *all but finitely many* pairs  $(q, n)$ .

Simultaneous extensiveness  $\implies$  Simultaneous  $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank  $\leq r$   
 $\implies$  Simultaneous Carlitz-extensiveness

## References & Acknowledgements

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- Thanks to **Carsten Rose** for IT support throughout the project.
- Thanks to the **SageMath community** for providing the mathematical software that made this work possible.

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<https://www.sagemath.org>

Thank you!

Questions?

## Counterexample: $\mathbb{F}_4/\mathbb{F}_2$ is not Carlitz-extensive

**Setup.**  $\mathbb{F}_{q^n} = \mathbb{F}_{2^2} = \mathbb{F}_4 = \mathbb{F}_2[x]/(x^2 + x + 1)$ ,  $\mathbb{F}_q = \mathbb{F}_2$ .  $\alpha$  root of  $x^2 + x + 1$ , so  $\alpha^2 = \alpha + 1$ .

- $|\mathbb{F}_4^\times| = 3$  is prime  $\implies$  primitive elements are  $\{\alpha, \alpha + 1\}$
- $x$  is a  $\gamma_z$ -generator  $\iff \{x, \gamma_z(x)\}$  is an  $\mathbb{F}_2$ -basis, where  $\gamma_z(x) = z \cdot x + x^2$

| $z$          | $\{\alpha, \gamma_z(\alpha)\}$ | $\{\alpha + 1, \gamma_z(\alpha + 1)\}$ |
|--------------|--------------------------------|----------------------------------------|
| 0            | $\{\alpha, \alpha + 1\}$ ✓     | $\{\alpha + 1, \alpha\}$ ✓             |
| 1            | $\{\alpha, 1\}$ ✓              | $\{\alpha + 1, 1\}$ ✓                  |
| $\alpha$     | $\{\alpha, 0\}$ ✗              | $\{\alpha + 1, \alpha + 1\}$ ✗         |
| $\alpha + 1$ | $\{\alpha, \alpha\}$ ✗         | $\{\alpha + 1, 0\}$ ✗                  |

**Conclusion:** For  $z \in \{\alpha, \alpha + 1\}$ , only 1 but no primitive element is a  $\gamma_z$ -generator

# Cyclotomic cosets and Frobenius compatibility

## Cyclotomic coset (modulo $q^n - 1$ )

$$C_k = \{k, kq, kq^2, \dots\} \pmod{q^n - 1}, \quad |C_k| \mid n.$$

For a primitive  $g \in \mathbb{F}_{q^n}$ , exponentiation turns  $C_k$  into the **Frobenius orbit** of  $g^k$ , and yields the minimal polynomial  $f_{g^k} = \prod_{j \in C_k} (x - g^j)$  easily.

## Lemma (Conjugation lemma)

For every  $z \in \mathbb{F}_{q^n}$  and  $i \in \mathbb{Z}$ ,  $\gamma_{\sigma^i(z)} = \sigma^i \circ \gamma_z \circ \sigma^{n-i}$ .

- If  $\alpha$  is a primitive  $\gamma_z$ -generator, then  $\sigma^i(\alpha)$  is a primitive  $\gamma_{\sigma^i(z)}$ -generator.
- **Reduction:** test only **one  $z$  per Frobenius orbit**.

# Projective space and $\mathbb{F}_q^\times$ -invariance

Set  $N = q^n - 1$  and  $Q = N/(q - 1)$ . The projective  $(n-1)$ -space:

$$\mathbb{P}_{\mathbb{F}_q}^{n-1} = \mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times \quad (\text{cyclic of order } Q).$$

A projective point  $[g^k]$  is **primitive** if it generates this quotient, i.e.  $\gcd(k, Q) = 1$ .

## Lemma ( $\mathbb{F}_q^\times$ -invariance)

Let  $z, x \in \mathbb{F}_{q^n}$  and  $\lambda \in \mathbb{F}_q^\times$ . Then

$$x \text{ is a } \gamma_z\text{-generator} \iff \lambda x \text{ is a } \gamma_z\text{-generator}.$$

**Reduction:** the  $\gamma_z$ -generator property depends only on the projective point  $[x]$ . Once a primitive projective point that is a  $\gamma_z$ -generator is found, some scalar multiple  $\lambda x$  is automatically a primitive  $\gamma_z$ -generator.

## A prerequisite: from individual to simultaneous cyclicity

The simultaneous extensiveness conjectures require a simultaneous (non-primitive) generator as hypothesis. When is this guaranteed?

### Conjecture (Individual $\Rightarrow$ simultaneous cyclicity)

Let  $(\tau_d)_{d|n, d < n}$  be a family of individually cyclic  $\mathbb{F}_{q^d}$ -linear endomorphisms of  $\mathbb{F}_{q^n}$ . Then a simultaneous generator exists, i.e., there is  $\alpha \in \mathbb{F}_{q^n}$  with

$$\mathbb{F}_{q^d}[\tau_d] \cdot \alpha = \mathbb{F}_{q^n} \quad \text{for all } d \mid n, d < n.$$

- $\tau_d = \sigma^d$ : Existence of completely normal elements proven by [Blessenohl–Johnsen \(1986\)](#).
- $\mathbb{F}_q[x]$ -Drinfeld special case  $\tau_d = \varphi^{(d)}$ : Fix  $z, c_1, \dots, c_r \in \mathbb{F}_{q^n}$ .

$$\forall d \mid n, d < n, \exists \text{ a } \varphi^{(d)}\text{-generator } \alpha \in \mathbb{F}_{q^n} \stackrel{?}{\implies} \exists \text{ a } \left( \varphi^{(d)} \right)_{d|n, d < n}\text{-generator } \alpha \in \mathbb{F}_{q^n}$$

- [Carlitz-special case](#) with  $\varphi^{(d)} = \gamma_z^{(d)}$ .